

1.4: Composition of Functions

Definition. (The Composition of Two Functions)

Let f and g be functions. Then the composition of g and f is the function $g \circ f$ defined by

$$(g \circ f)(x) = g(f(x))$$

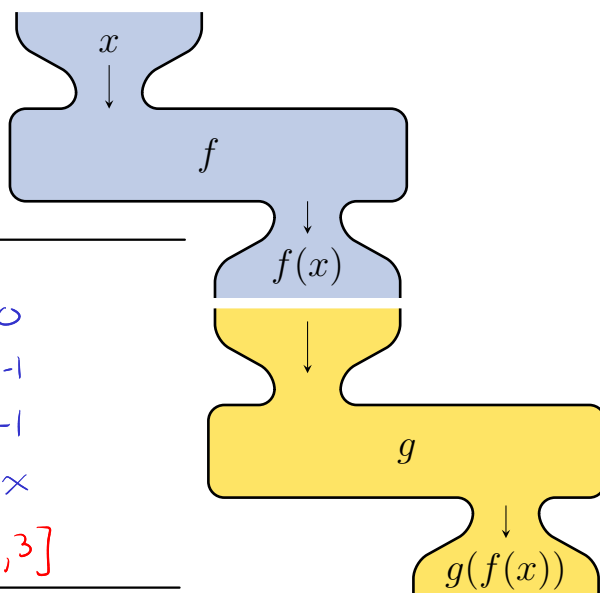
The domain of $g \circ f$ is the set of all x is the domain of f such that $f(x)$ lies in the domain of g .

Example. Let $f(x) = \sqrt{x+1}$ and $g(x) = 4-x$. Find the domain of the following:
 $[-1, \infty)$ $(-\infty, \infty)$

$$\begin{aligned} g(f(x)) &= 4 - f(x) \\ &= 4 - \sqrt{x+1} \rightarrow [-1, \infty) \end{aligned}$$

$$\begin{aligned} f(g(x)) &= \sqrt{g(x) + 1} \rightarrow g(x) + 1 \geq 0 \\ &= \sqrt{4 - x + 1} \quad \begin{aligned} g(x) &\geq -1 \\ 4 - x &\geq -1 \\ 3 &\geq x \end{aligned} \\ &= \sqrt{3 - x} \rightarrow (-\infty, 3] \end{aligned}$$

$$\begin{aligned} f(f(x)) &= \sqrt{f(x) + 1} \rightarrow f(x) + 1 \geq 0 \\ &= \sqrt{\sqrt{x+1} + 1} \quad \begin{aligned} f(x) &\geq -1 \\ \sqrt{x+1} &\geq -1 \\ \text{Always true} \end{aligned} \\ &\rightarrow [-1, \infty) \end{aligned}$$



Example. For the following, find functions $g(x)$ and $h(x)$ so that $f(x) = g(h(x))$:

$$f(x) = \frac{1}{x+2}$$

$$g(h) = \frac{1}{h}$$

$$h(x) = x+2$$

$$f(x) = \sqrt{x-4}$$

$$g(h) = \sqrt{h}$$

$$h(x) = x-4$$

$$f(x) = (x^2 + 3x - 8)^4$$

$$g(h) = h^4$$

$$h(x) = x^2 + 3x - 8$$

$$f(x) = \frac{x-3}{\sqrt{-(x-3)}}$$

$$g(h) = \frac{h}{\sqrt{-h}}$$

$$h(x) = x-3$$